<u>CS4782 Final Project Proposal</u>

Member 1: Nicholas Josephson - noj5
Member 2: Julius Samwer — jcs557
Member 3: Kevin Rodriguez — kjr64
Member 4: Tudor Braicu — tb574

1. Paper selection:

Title:

- Cold Diffusion: Inverting Arbitrary Image Transforms Without Noise Diffusion (https://arxiv.org/abs/2208.09392)

Authors:

- Arpit Bansal, Eitan Borgnia, Hong-Min Chu, Jie S. Li, Hamid Kazemi, Furong Huang, Micah Goldblum, Jonas Geiping, Tom Goldstein

Venue/status:

- arXiv preprint (Aug 2022); later published at NeurIPS 2023

Brief summary of the chosen paper

- The paper talks about building diffusion models on arbitrary transformations, including deterministic transformations or 'cold' diffusions. In doing so, the paper tries to answer the question of what effect the transformation we apply has on our output, highlighting the unimportance of Gaussian noise in diffusion models.

Brief justification of why you chose this paper

- We chose this paper in order to explore the principles behind diffusion models. We were interested by the idea of taking out one of the core principles of diffusion models, yet still achieving good results, which we hope can provide insight into how to select degradation functions. Furthermore, none of us is that familiar with diffusion models, and we take this as an opportunity to learn more about them.

2. Data

The paper's experiments are built on standard public datasets:

- MNIST, CIFAR-10, CelebA (diffusion)

For generation experiments, they also use: AFHQ (animal faces)

3. Result Selection

First, we would like to replicate the result that demonstrates that Algorithm 2 sampling produces more realistic outputs (FID), even if the pixelwise metrics worsen. We will do this by comparing direct reconstruction to our Algorithm 2 reconstruction on the test set.

Second, we'd also like to demonstrate that the second algorithm is also more stable than Algorithm 1 on deblurring. This will involve showing that Algorithm 2 doesn't drift or degrade the sample quality as the other one does.

4. Re-implementation Plan (Describe architecture, method, and metrics)

The paper uses various models for the different transformations, but they all tend to agree on a U-Net backbone: a set of ResNet blocks, decoders, encoders, and skip layers. The general architecture is then composed of Input → U-Net layers downsample → U-Net layers upsample → Output. From our research, we suspect an architecture of the form:

<u>Input</u>: (x_t, t)

↓ Convolution layer
↓ ResBlock
↓ Max pool
↓ ResBlock
↓ Max pool
↓ ResBlock
…
↓ ResBlock with Attention
…
↓ Upsample
↓ ResBlock + Skip Connection
↓ Attention
↓ Upsample
↓ ResBlock

<u>Output</u>: predicted clean image x(hat) _0 (same shape as input image)

Degradation: $xt=D(x0,t)$xt=D(x0,t) with $D(x,0)=x$D(x,0)=x
diffusion
Restoration network: learn $R(xt,t)≈x0$R(xt,t)≈x0 by minimizing an ℓ1ℓ1 reconstruction loss
Diffusion

Sampling:

- Baseline: Algorithm 1 ("naive"): *x*^0=*R*(*xs*,*s*)x^0=R(xs,s), then *xs*−1=*D*(*x*^0,*s*−1)xs−1=D(x^0,s−1)
  diffusion
- Proposed: Algorithm 2 ("improved cold sampling"): *xs*−1=*xs*−*D*(*x*^0,*s*)+*D*(*x*^0,*s*−1)xs−1=xs−D(x^0,s)+D(x^0,s−1)

Method:

To replicate this paper, we will start by first attempting to get close to their results for a standard diffusion model that uses Gaussian degradation. This should ensure we have the correct environment setup, such as using the right metrics and dataset. We can then implement the generalized diffusion model outlined, starting by picking a deterministic degradation function, and then training our network to learn restoration. We can then use this network and sample with both Algorithms 1 and 2 to get our output images, comparing the metrics and benefits of each. We can then compare these results to standard diffusion, hopefully extending it to unconditional generation.

Metrics :

The paper utilizes FID **(** Closeness to a set of real images that represent a "ground truth") vs held-out test set as its main evaluation metric. Also employs RMSE, SSIM, and PSNR where applicable. These evaluation metrics capture the tradeoff between visually realistic images and the accuracy when matching exact ground truth pixels.

Compute :
Deblurring: 700,000 gradient steps (Adam LR 2 x 10^{-5}, batch size 32 (+ grad accumulation)),
Inpainting: 60,000 gradient steps (Adam LR 2 x 10^{-5}, batch size 64 (+ grad accumulation)),
Super-resolution: 700,000 iterations (Adam LR 2 x 10^{-5}, batch size 32 (+ grad accumulation))
Snowification: 700,000 gradient steps (Adam LR 2 x 10^{-5}, batch size 32 (+ grad accumulation)
For all runs, we use gradient accumulation every 2 steps, and EMA model selection with decay 0.995 updated every 10 steps.

Feasible without matching their full compute :
Reduce steps (e.g., 50k–200k) and show the trend:, very manageable on 1 GPU (Google Coop Pro)